

BARRE KNAPP, VT, USA

WMO: 726145

Lat: 44.204N

Lon: 72.562W

Elev: 1126

StdP: 14.11

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 94705

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.3	-5.2	-18.1	2.1	-9.1	-13.1	2.8	-3.4	23.2	20.5	20.6	19.6	5.0	320	0.556

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.2	84.9	69.8	82.1	68.2	79.6	67.0	72.5	80.9	70.8	78.7	69.1	76.4	7.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.8	114.3	76.6	68.1	107.6	75.1	66.4	101.3	73.5	36.8	81.1	35.2	78.5	33.9	76.6	86.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.5	17.5	15.6	-18.2	89.5	5.5	2.7	-22.2	91.4	-25.4	93.0	-28.5	94.5	-32.5	96.4		
(5)			-18.5	75.9	5.5	2.8	-22.5	77.9	-25.7	79.5	-28.7	81.1	-32.7	83.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.6	17.3	19.9	28.6	41.7	54.0	62.7	67.2	65.5	58.4	46.6	35.1	24.0
	DBStd	19.65	13.01	11.20	10.64	8.90	7.69	6.65	5.13	5.40	7.28	8.14	9.52	10.63
	HDD50	4284	1013	842	666	280	43	1	0	0	12	167	454	806
	HDD65	8094	1478	1262	1127	702	349	121	35	61	220	572	896	1271
	CDD50	1933	1	0	3	30	169	382	534	481	265	60	7	1
	CDD65	268	0	0	0	1	10	51	104	78	23	1	0	0
	CDH74	2396	0	0	3	31	143	523	881	605	202	8	0	0
	CDH80	492	0	0	0	5	23	127	194	106	37	0	0	0
	WSAvg	5.8	6.3	6.6	7.1	7.0	6.4	5.3	4.8	4.3	4.8	5.6	6.2	5.8
	PrecAvg	34.4	2.1	2.0	2.3	2.4	3.2	3.4	3.1	3.8	3.1	2.9	3.1	2.8
(15) Precipitation	PrecMax	47.4	6.9	6.3	3.7	4.9	7.7	7.4	5.9	10.7	7.1	5.5	5.7	6.3
	PrecMin	24.6	0.2	0.4	0.4	0.7	0.4	1.4	1.1	1.3	0.7	0.2	1.3	1.1
	PrecStd	5.9	1.5	1.1	0.9	1.1	1.7	1.5	1.1	2.0	1.5	1.4	1.2	1.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.9	52.6	66.1	78.3	83.4	88.2	88.5	86.9	84.9	74.9	65.0	54.0
		MCWB	48.0	45.4	54.4	59.7	65.8	70.6	73.1	71.4	69.6	62.0	56.9	48.2
	2%	DB	45.6	45.8	54.6	69.9	78.3	83.6	84.8	82.8	79.3	69.2	59.3	47.1
		MCWB	42.1	39.8	45.5	53.9	61.6	68.4	70.6	69.1	67.5	59.2	53.1	43.4
	5%	DB	39.1	40.1	48.5	63.5	73.5	80.0	81.9	80.2	75.5	64.6	54.6	42.4
		MCWB	36.0	35.5	41.1	50.6	59.3	66.4	68.8	67.9	65.0	56.9	49.4	39.2
	10%	DB	34.8	36.0	43.5	57.2	69.5	76.3	79.2	77.4	71.6	60.2	49.6	37.9
		MCWB	32.5	32.5	37.6	46.6	57.4	64.3	67.4	66.7	63.1	54.0	44.6	35.0
	0.4%	WB	49.7	47.7	54.5	61.7	68.0	73.6	75.3	73.9	72.2	65.6	59.0	49.7
		MCDB	52.2	50.4	63.8	73.3	79.1	82.7	84.5	82.3	80.8	70.2	63.2	53.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	41.5	40.4	47.2	56.5	65.0	71.1	72.9	71.4	69.3	61.4	54.4	43.8
		MCDB	44.6	44.8	53.2	67.0	73.7	80.6	81.3	79.0	76.7	66.3	58.6	46.7
	5%	WB	36.2	36.1	41.9	52.4	62.3	68.8	71.1	69.8	67.1	58.0	50.0	39.3
		MCDB	38.6	39.6	47.2	60.7	70.3	76.6	78.7	76.9	73.5	63.9	54.2	42.0
	10%	WB	32.7	32.9	38.3	48.0	59.4	66.4	69.3	68.2	64.6	54.6	45.0	35.4
		MCDB	34.9	35.6	43.0	56.9	67.1	73.4	76.2	75.0	69.9	59.1	48.9	37.7
(35) Mean Daily Temperature Range	5% DB	MDBR	16.6	18.0	17.5	19.4	20.8	20.4	20.2	20.4	20.4	17.8	15.1	14.4
		MCDBR	19.2	20.8	24.1	31.2	28.1	26.0	23.6	23.3	24.7	25.0	22.6	17.6
		MCWBR	16.6	16.4	16.4	17.8	14.6	13.0	11.4	11.7	13.6	15.8	17.0	15.2
	5% WB	MCDBR	17.5	19.8	21.4	26.4	22.8	21.7	20.2	19.8	20.5	21.5	20.3	17.1
		MCWBR	15.6	16.3	15.9	16.9	13.1	11.9	10.7	10.8	12.3	15.1	16.9	15.4
(40) Clear-Sky Solar Irradiance	taub		0.271	0.280	0.316	0.346	0.368	0.419	0.430	0.403	0.357	0.326	0.302	0.272
	taud		2.382	2.331	2.361	2.401	2.394	2.248	2.225	2.291	2.413	2.511	2.513	2.453
	Ebn at Noon		273	290	292	291	286	271	266	269	275	271	261	262
	Edh at Noon		24	30	34	36	37	43	43	39	32	25	21	20
(44) All-Sky Solar Radiation	RadAvg		459	734	1091	1360	1575	1710	1753	1584	1275	757	481	342
	RadStd		53	75	71	167	176	155	130	81	95	82	61	31

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+128	N/A

Nomenclature: See separate page